

Set-1 Scenario Based

1. Image Class:

- Add Dropout and BatchNorm
- Use data augmentation
- Apply early stopping
- Use a pre-trained model for this transfer learning

2. ANN fails:

- ANN may not capture non linear features in images
- Replace ANN with CNN for image based tasks
- Normalize input images
- Use ReLU activation instead of sigmoid

3. CNN Model Prediction:

- Dataset is likely imbalanced with more mask images
- Use class weights or resample the dataset
- Monitor confusion matrix and recall, precision and accuracy

4. YOLO:

- Labeling format might be incorrect

- Use correct YOLO
format:(class_id,x_center,y_center,width,height)
normalized to image size
- Check dataset with label visualization like roboflow

5.Face Mask detection:

- Training images were likely frontal and clear
webcam has varied angles and lighting
- Use data augmentation
- Collect real time webcam data and fine tune the
model

6.Object detector:

- Model not trained on night/low night images
- Train using night mode data or augment
brightness
- Apply Histogram equalization or CLAHE
preprocessing